

CM1605 Web Technology

Introduction to Web Technologies & HTML I

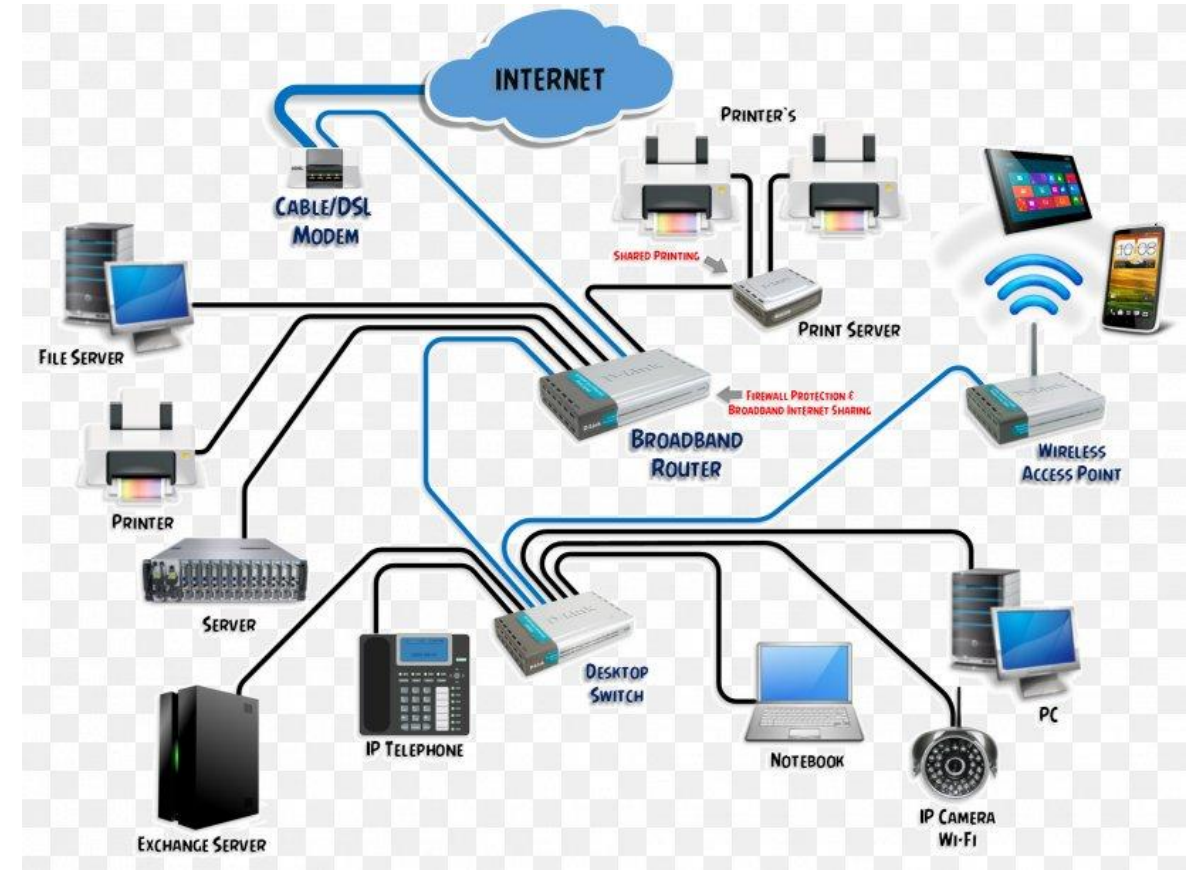
Week 1 | Janani Harischandra

Learning Outcomes

- Covers LO1 and LO2 for Module
- On completion of this lecture, students are expected to be able to:
 - Define a computer network, internet, internet protocols, internet addressing, domain name servers, URL and URI,
 - Describe different web application architectures, web servers, web browsers and web search engines and give real world examples of them.
 - Define and understand SGML, XML, HTML terminologies.
 - Understand and describe HTML page structures.
 - Apply hypertext concepts in making hypertext documents using the current HTML standard with different Text formatting tags, Logical tags, Links, Lists.

Computer Networks

- A computer network can be defined as a set of computers (nodes) and different other computing hardware devices connected through communication channels (cabling/wireless) to facilitate communication and resource sharing among a wide range of users.



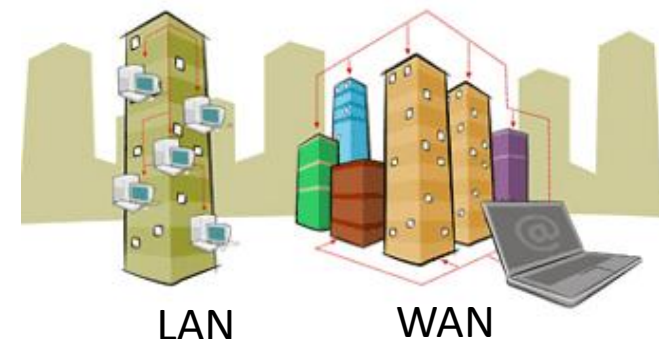
Computer Network Types

- **LAN (Local Area Network)**

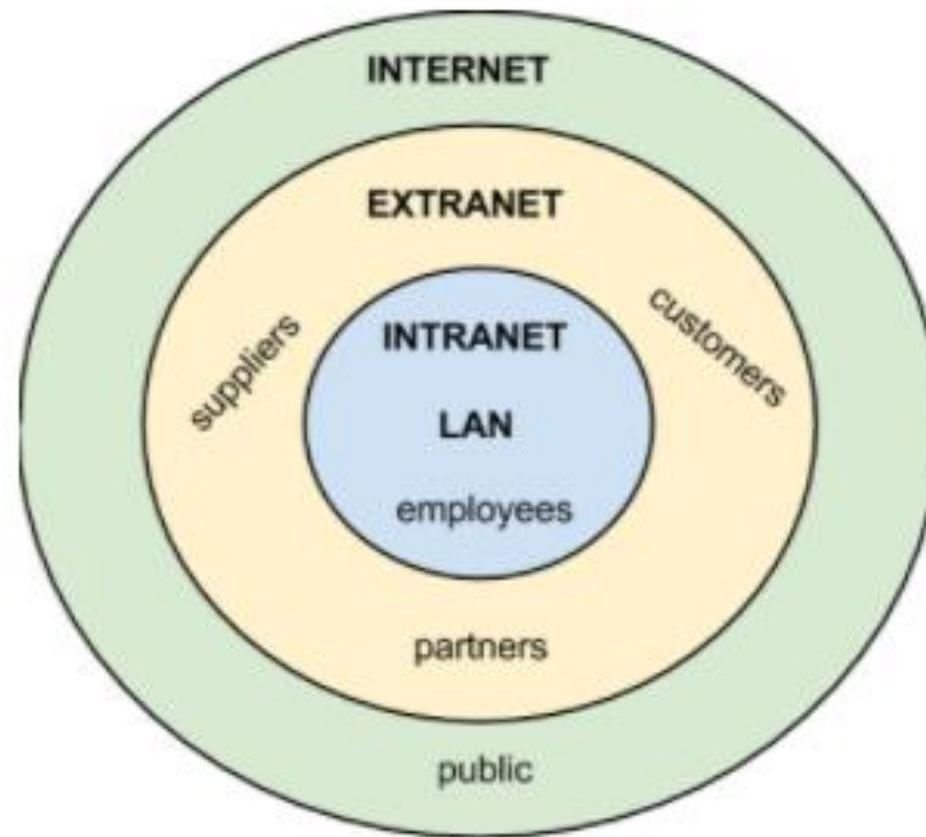
- Is a group of computers connected to each other in a small area such as building, office which connects two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and Ethernet cables.

- **WAN (Wide Area Network)**

- is a network that extends over a large geographical area such as states or countries connected through a telephone line, fibre optic cable or satellite links.
- The **internet** is one of the biggest WAN in the world.



Scope of business networks



History of Internet

- In late 1960's **ARPANET** was invented by a group of MIT graduate students which later evolved in to Internet today.
- ARPANET primary goals was to allow multiple users to send and receive information simultaneously over the communication paths(eg: phone lines).

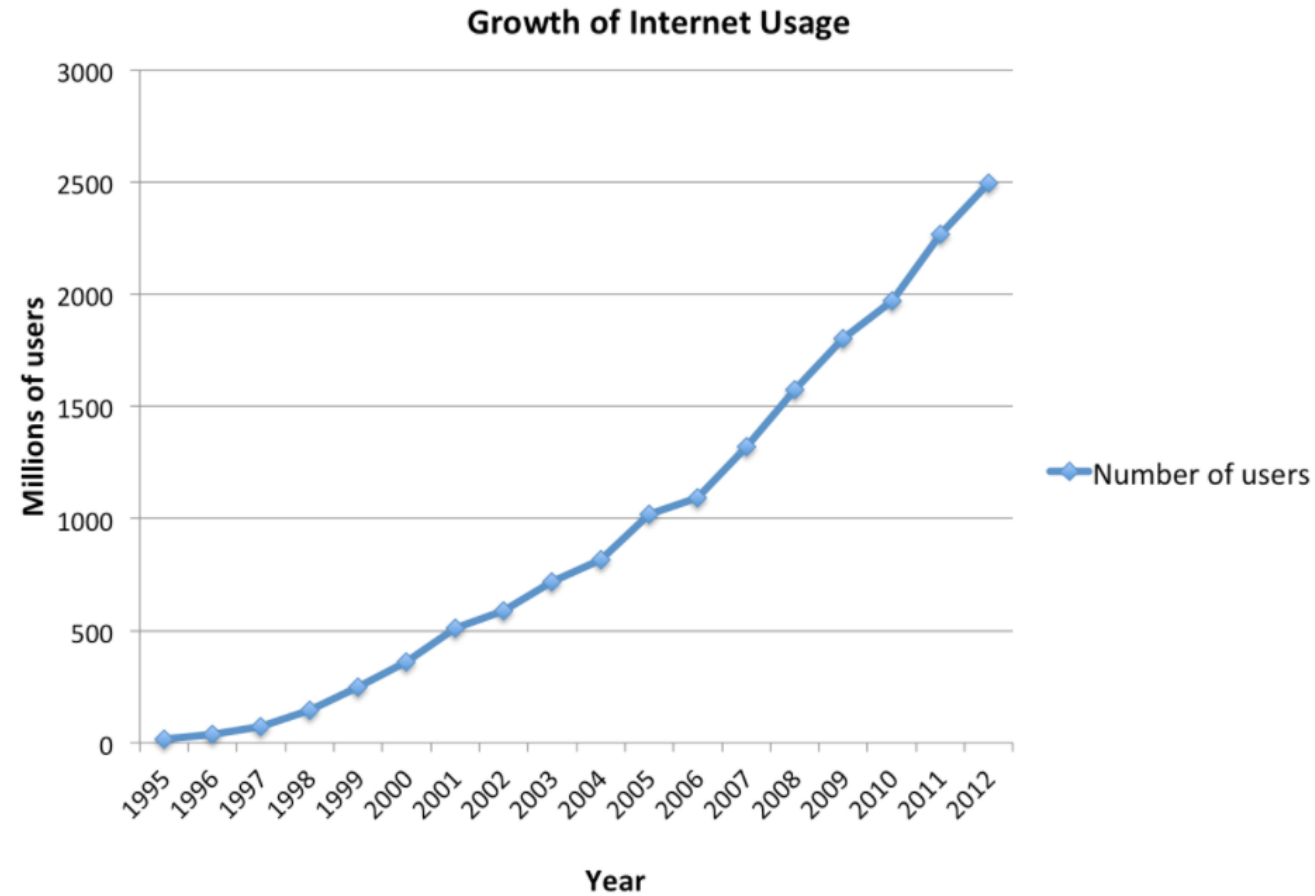
History - <https://www.computerhistory.org/internethistory/>

Internet

- The Internet is a global wide area network that connects computer systems across the world which exchange data cooperating with each other.
- Read More About
- Internet - https://www.youtube.com/watch?v=7_LPdttKXPc
<https://www.youtube.com/watch?v=x3c1ih2NJEg>



Growth of Internet



Growth of internet usage, 1995–2012 .Data taken from InternetWorldStats.com.

World Wide Web Consortium (W3C)

- In 1994, [World Wide Web Consortium \(W3C\)](#) was founded to develop non proprietary technologies for the web.
- Primary to goal is to W3C's to make the web universally accessible – regardless of ability , language or culture.
- Web technologies standardized by the W3C are called **Recommendations**.

Internet Communication

- Protocols
- Addressing
 - IP Address
 - Domain Name
- URL

Protocols

- Protocols are set of rules for routing and addressing packets of data so that they can travel across networks and arrive at the correct destination.
- Examples :
 - HTTP - Hypertext Transfer Protocol
 - HTTPS - Secure Hypertext Transfer Protocol
 - TCP/IP – Transmission Control Protocol / Internet Protocol
 - FTP – File Transfer Protocol

Internet Addressing

- Every computer / device connected to the Internet has a unique identity.
- The main objective of internet addressing is to locate each device/host/computer by a unique address.
- There are different types of addressing introduced in order to achieve the above purpose.
- Types of internet addressing
 - Internet Protocol address (IP)
 - Domain Name Server address (DNS)

Internet Protocol Address (IP)

- Every machine/device on network has a unique identifying number, called an IP Address.
(router,smartphone,etc)



199.181.132.249
199.181.132.250

- A typical IP address looks like this(ipv4):
- The format of an IP address is a 32-bit numeric address with 12 digits written as four numbers separated by periods/full stops where each number can be a number between zero and 255.
- When constructing a network, IP addresses can be assigned randomly provided that each IP address is a unique address, this uniqueness is essential to avoid sending data to multiple destinations or to the wrong destination.

Brainstorm

- What is Ipv6? And Why?

Domain Name Server (DNS)

- Friendly addressing (DNS), introduced to help users remembering (or even guessing) the name of a host.

Advantage:

- Numeric addresses are difficult to work with
- Extremely difficult to remember as humans

Eg:

- `http://10.0.7.200:8080` Vs. `http://moodle.iit.lk`

Disadvantage:

- Unable to generate meaningful and unique names for all devices in the Internet

Domain Levels

- Domain names are organized in a hierarchical manner.
 - Top Level Domain (TLD) : Last part of the Domain name
 - e.g. : **org** is TLD of `www.wikipedia.org`
 - Second Level Domain: Name directly to the left of TLD
 - e.g. : **wikipedia** is SLD of `www.wikipedia.org`
 - Third Level Domain: Name directly to the left of second level Domain
 - e.g. : **mail** is SLD of `mail.google.com`

Top Level Domain Types

- Commercial Organizations com
- Educational Institute edu
- Government Institute gov
- Military Groups mil
- Major network support centers net
- Non profit Organizations org
- International Organizations int

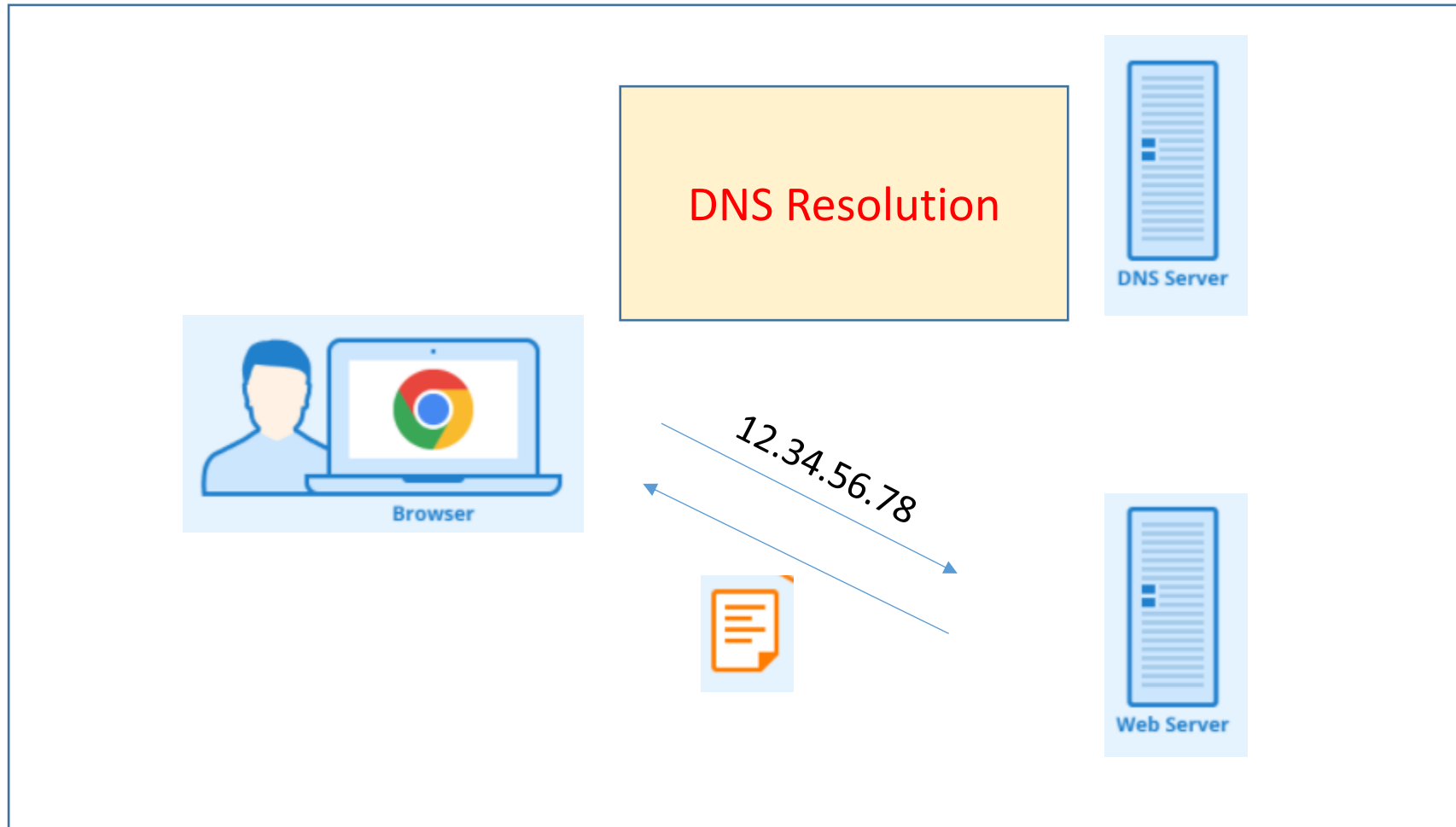
Domain Name System(DNS)

- The Domain Name System (DNS) works as the phonebook of the Internet.
- DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4) and IPv6.
- DNS is a database system that translates a computer's fully qualified domain name into an IP address.
- The DNS functions by converting the domain name into an IP address character set, via a Domain Name System server.

Domain Name Resolution

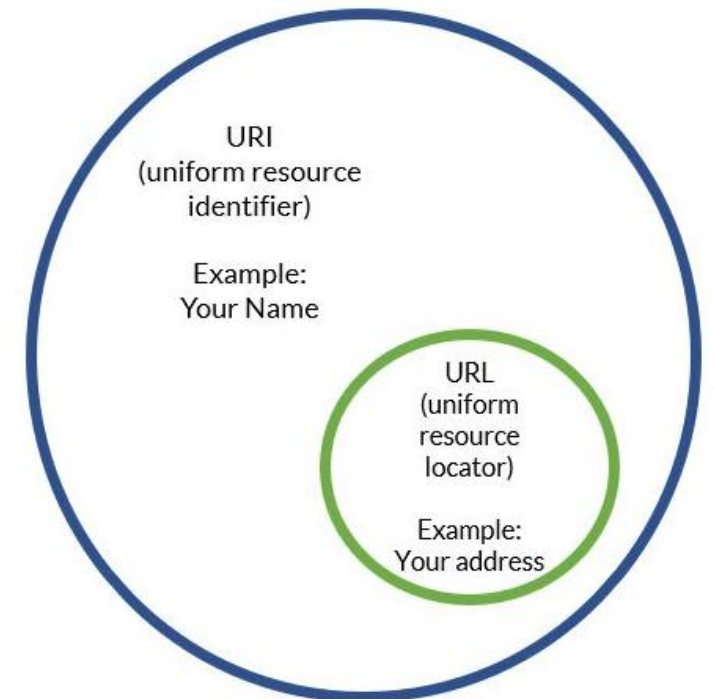
- When you type web address in your browser eg: google.com, bbc.co.uk, ebay.com.
- Your computer contacts DNS servers which returns the IP address associated with the requested domain name.
- The unique number that the DNS server returns to your computer allows your browser to contact the web server that hosts the website requested.
- Translating the name into the IP address is called “resolving the domain name” or “DNS resolution”.
- If a name server in the local domain cannot resolve a client's request, it queries other servers to locate a server that can.

Domain Name Resolution



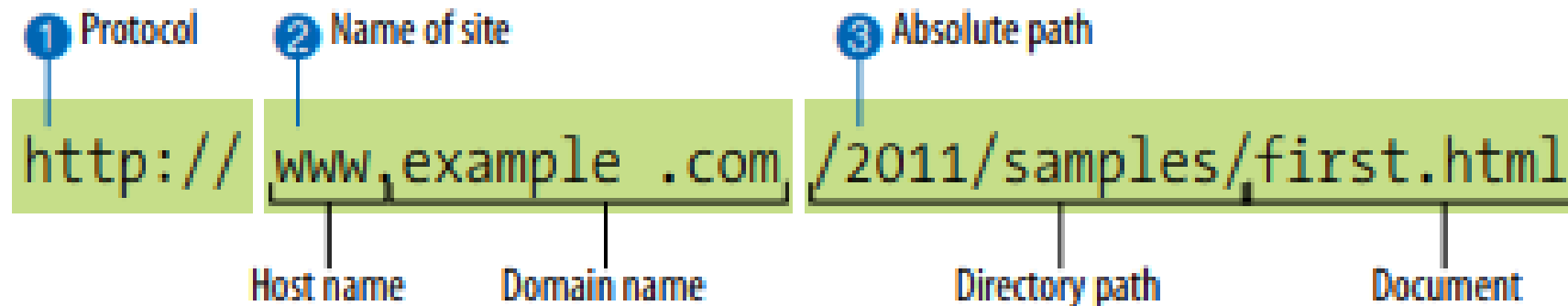
Uniform Resource Identifier (URI)

- URI can be a name, locator, or both for resource in the web where a URL is just the locator.
- URLs are a subset of URIs.
- All URLs are URIs. It doesn't work the opposite way though.

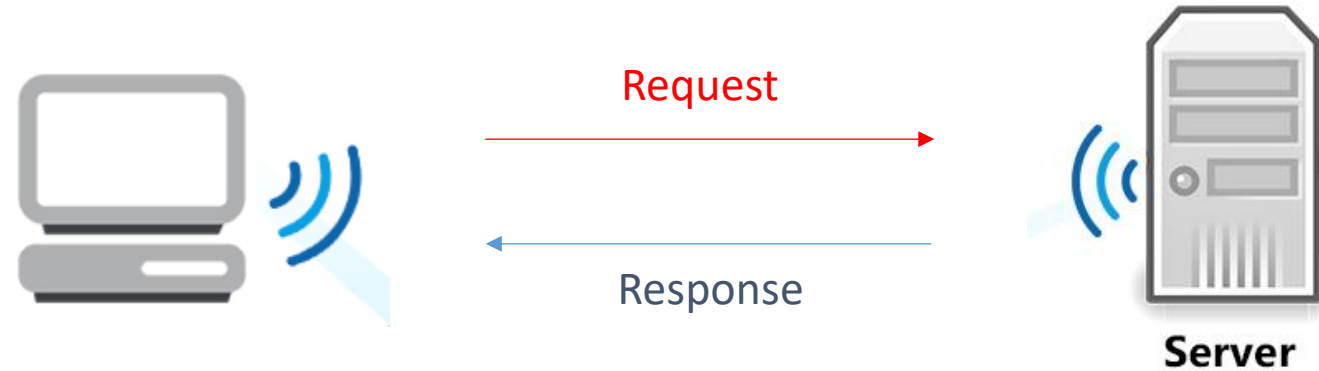


Web Page Addresses (URLS)

- Every web page and resource on the web has its own special address called a URL.
- URL stands for Uniform Resource Locator.
- This is the web address that you would type into a browser if you wanted to visit that specific page.



Client Sever Architecture



Client/server describes the relationship between two computer programs in which one program, the client, makes a service request from another program, the server, which fulfils the request

Server

- Computer program or machine that provide services to (Web Services or FTP services) to other programs or users(machines).

Client

- Computer program or machine that request the services from servers.

Types of Client Server Architecture

1-Tier
Architecture

2-Tier
Architecture

3-Tier
Architecture

N-Tier
Architecture

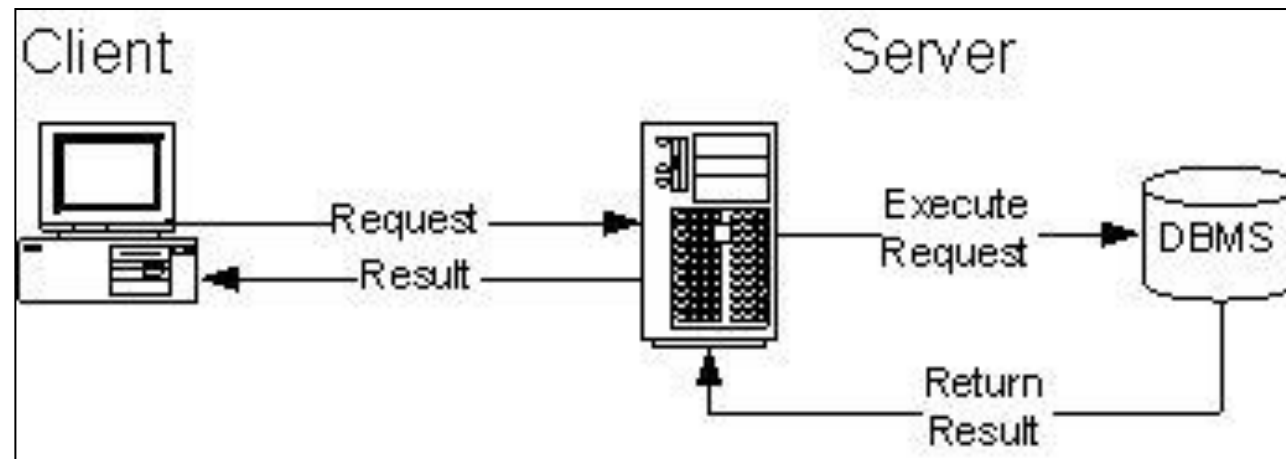
1- Tier Architecture

- It is the simplest one as it is equivalent to running the application on the personal computer. All of the required components for an application to run are on a single application or server including Client, Server, Database.



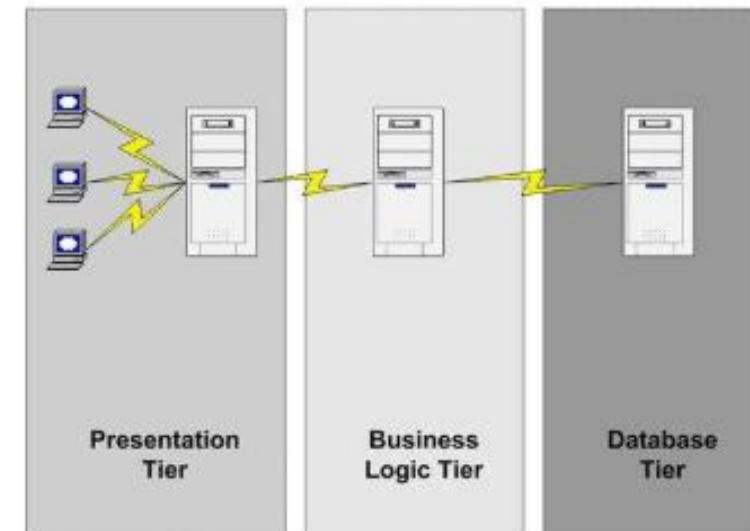
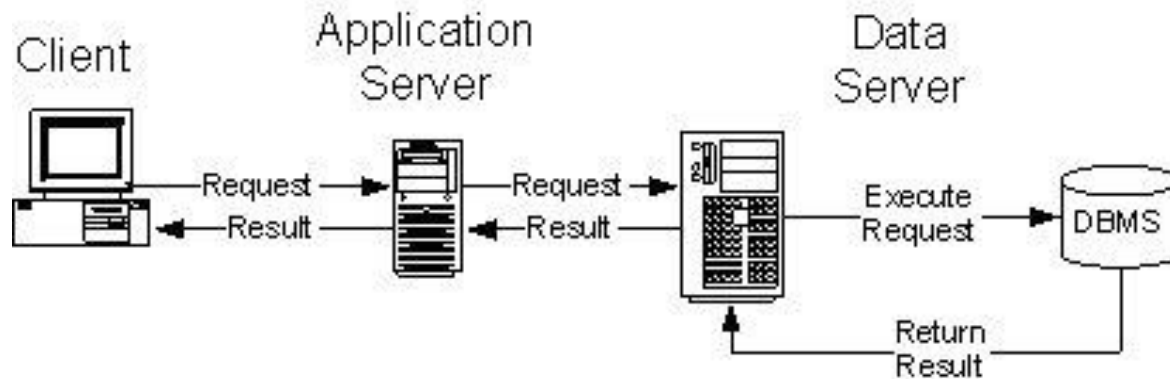
2-Tier Architecture

- A two-way interaction in a client/server environment, in which the user interface is stored in the client and the data are stored in the server.
- The **application logic** can be in either the client or the server.



3-Tier Architecture

- A three-way interaction in a client/server environment, in which the user interface is stored in the client, the bulk of the business application logic is stored in one or more servers, and the data are stored in database server.
- The 3-tier architecture is split into 3 parts, namely, The presentation layer (Client Tier), Application layer (Business Tier) and Database layer (Data Tier).
- The Client system manages Presentation layer; the Application server takes care of the Application layer, and the Server system supervises Database layer.



N-Tier Architecture

- **The N-tier (multi-tier) architecture** is an industry-proven software architecture model.
- There are different types of N-Tier Architectures such as **3-tier Architecture, 2-Tier Architecture and 1- Tier Architecture.**
- It is suitable to support enterprise level client-server applications by providing solutions to scalability, security, fault tolerance, reusability, and maintainability.

- Read More : <https://www.guru99.com/n-tier-architecture-system-concepts-tips.html>

Web Browsers

- Web browsers are software programs that allow users to access the web's rich content.
- Information Resources : Images, videos, and other piece of content. Hyperlinks present in resources enable users to easily navigate their browsers to related resources.
- People use web browsers for different purposes such business, personal to access tremendous amounts of information available on the web.
- The popular browsers today are Google Chrome, Firefox, Safari, Edge

Web Browser Usages

- Read More : <https://www.w3schools.com/browsers/default.asp>

2020	<u>Chrome</u>	<u>Edge/IE</u>	<u>Firefox</u>	<u>Safari</u>	<u>Opera</u>
May	80.7 %	3.5 %	8.5 %	4.1 %	1.6 %
April	80.7 %	3.4 %	8.6 %	4.2 %	1.5 %
March	81.4 %	3.5 %	8.7 %	3.7 %	1.3 %
February	82.0 %	3.4 %	8.7 %	3.4 %	1.2 %
January	81.9 %	3.0 %	9.1 %	3.3 %	1.3 %

Web Servers

- A **web server**: computers on the Internet that host websites, serving pages to viewers upon request.
- Each web server has a **unique address (IP address) and domain name** and Provide services such as **Web hosting**.
- Any computer can be turned into a Web server by installing server software and connecting the machine to the Internet.
- Read more: https://w3techs.com/technologies/overview/web_server

Web Search Engines

- A web search engines can be named as a specialized tool which enables users to locate/explore information on WWW in a systematic way.
- The search results of a user query are often returned as a list.
- The list may consist of web pages, images, and other types of files.
- Some popular web search engines – Google, Yahoo, bling, msn, Ask.



Further Reading

- Deitel, P.J., and Deitel, H.M. (2005), *Internet and World Wide Web How to program*, Pearson.
- Duckett, J. (2011), *HTML & CSS, Design and Build Websites*, Wiley.
- Robinson, J . (2018), *Learning Web Design, A Beginner's Guide to HTML,CSS, JavaScript, And Web Graphics*, O'Reilly.

Content

- Client and Server Side Technologies
- Markup Language
- HTML and XML
- HTML 5
- HTML Terminologies
- Basic HTML page structure
- Basic HTML tags
- Text formatting tags
- Logical tags
- List tags
- HTML Links

Client Side Techniques for Web Development

HTML

- Hyper Text Markup Language used to create static web pages.

CSS

- Cascade Style Sheets used to define styles for HTML pages.

Java Scripts

- A scripting language that enables creating dynamically updating content, control multimedia, animate images.

XML

- Extensible Markup Language used to store and transport data.

Server Side Techniques for Web Development

PHP

- Personal Home Page is used to create dynamic and interactive web pages.

JSP

- Java Server Pages is a Java standard technology used to create dynamic data driven web pages.

ASP.Net

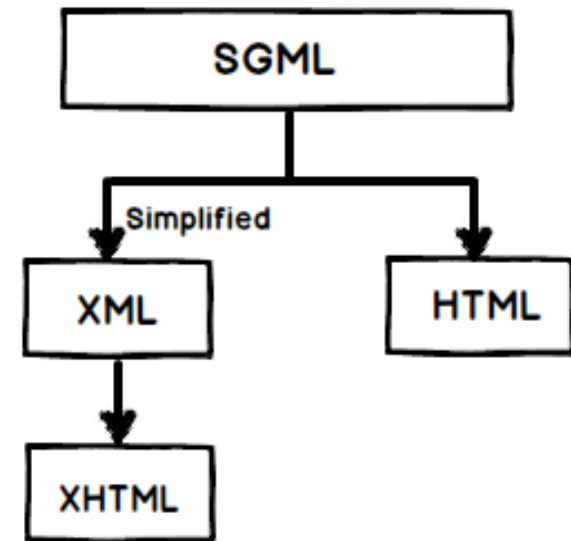
- A new framework for creating web applications.

Python

- Python offers many frameworks for web application development

Markup Language (SGML)

- SGML stands for Standard Generalized Markup Language.
- It is a standard for how to specify a document markup language or tag set.
- SGML is based on the idea that documents have structural and other semantic elements.
- Developed and standardized by the International Organization for Standards (ISO) in 1986.



HTML

- HTML (Hyper Text Markup Language) is the standard markup language for Web pages.
- It is the standard markup language for creating Web pages.
- HTML elements tell the browser how to display the content.
- HTML web pages are text documents.
- HTML uses tags (characters that sit inside angled brackets) to give the information they surround special meaning.
- Tags are often referred to as elements.

HTML History

Year	Version
1989	Tim Berners-Lee invented www
1991	Tim Berners-Lee invented HTML
1993	Dave Raggett drafted HTML+
1995	HTML Working Group defined HTML 2.0
1997	W3C Recommendation: HTML 3.2
1999	W3C Recommendation: HTML 4.01
2000	W3C Recommendation: XHTML 1.0
2008	WHATWG HTML5 First Public Draft
2012	<u>WHATWG HTML5 Living Standard</u>
2014	<u>W3C Recommendation: HTML5</u>
2016	W3C Candidate Recommendation: HTML 5.1
2017	<u>W3C Recommendation: HTML5.1 2nd Edition</u>
2017	<u>W3C Recommendation: HTML5.2</u>

HTML 5

- HTML5 is the newest version of HTML.
- It incorporates all features from earlier versions of HTML, including the stricter XHTML.
- Read more on HTML 5
 - <https://html.spec.whatwg.org/#is-this-html5?>
 - <https://www.w3.org/TR/2008/WD-html5-20080122/>



Why HTML 5?

- It promotes cross platforms.
- Improves semantic definitions.
- Consistency across of multiple browsers.
- A better user experience with a wider range of design and presentation tools across media types.
- Reduce the need for external plugins and scripts to show website content.
- Handle web documents errors in a better and more consistent fashion.

XML

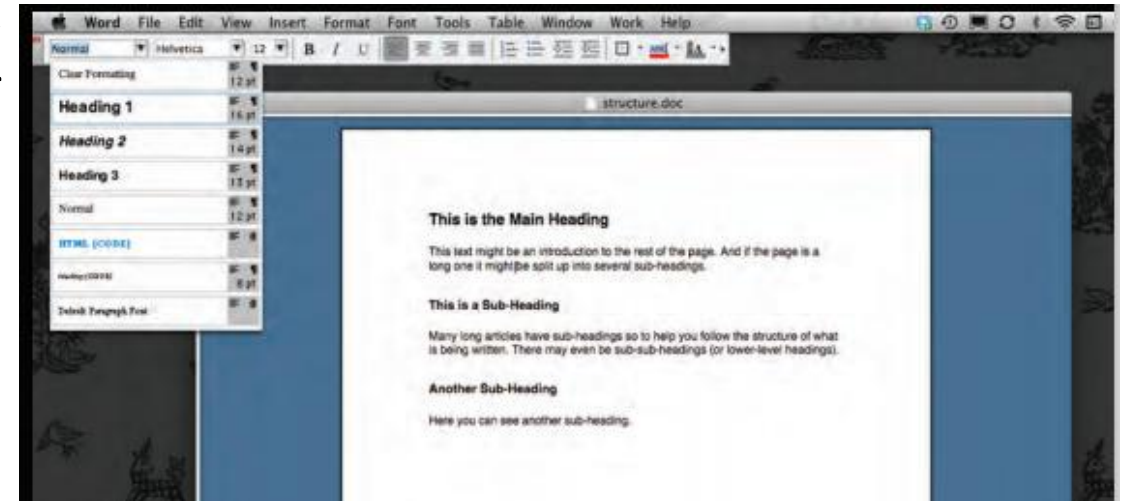
- XML is a software- and hardware-independent tool for storing and transporting data.
- XML has following features
 - XML stands for Extensible Markup Language
 - XML is a markup language much like HTML
 - XML was designed to store and transport data
 - XML was designed to be self-descriptive
 - XML is a W3C Recommendation

Difference (HTML & XML)

- XML and HTML were designed with different goals:
 - XML was designed to carry data - with focus on what data is
 - HTML was designed to display data - with focus on how data looks
 - XML tags are not predefined like HTML tags are

Structuring Word Documents

- The use of headings and subheadings in any document often reflects a hierarchy of information.
- Eg: document might start with a large heading, followed by an introduction or the most important information.
- This might be expanded upon under subheadings lower down on the page.
- In the browser window you can see a web page that features exactly the same content as the Word document.



Understanding the Document Structures



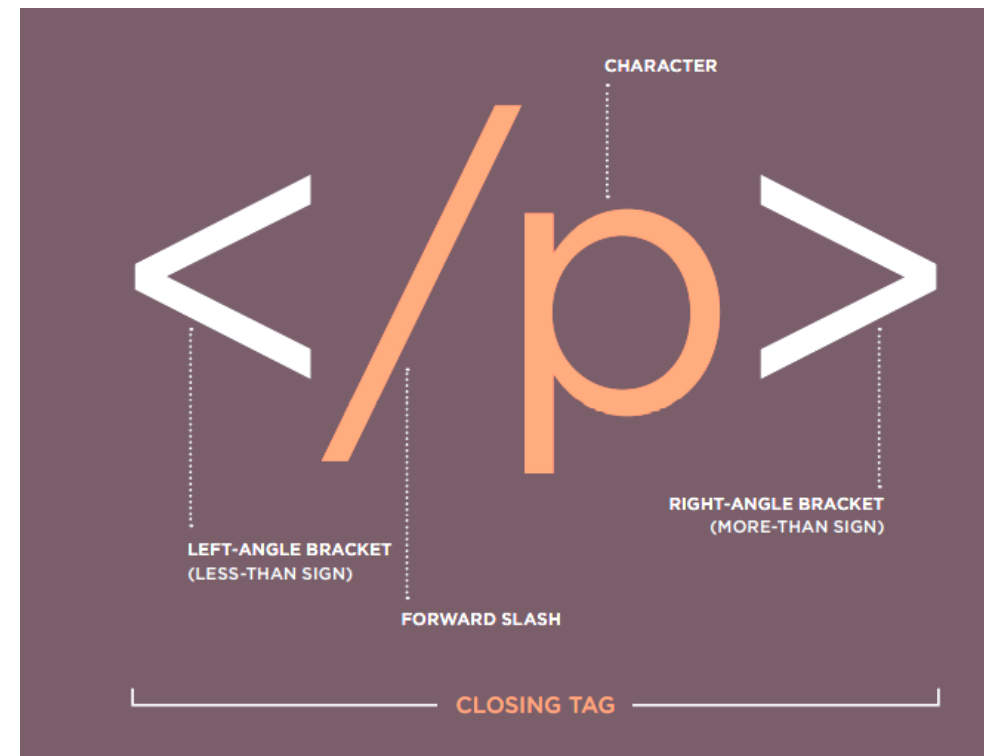
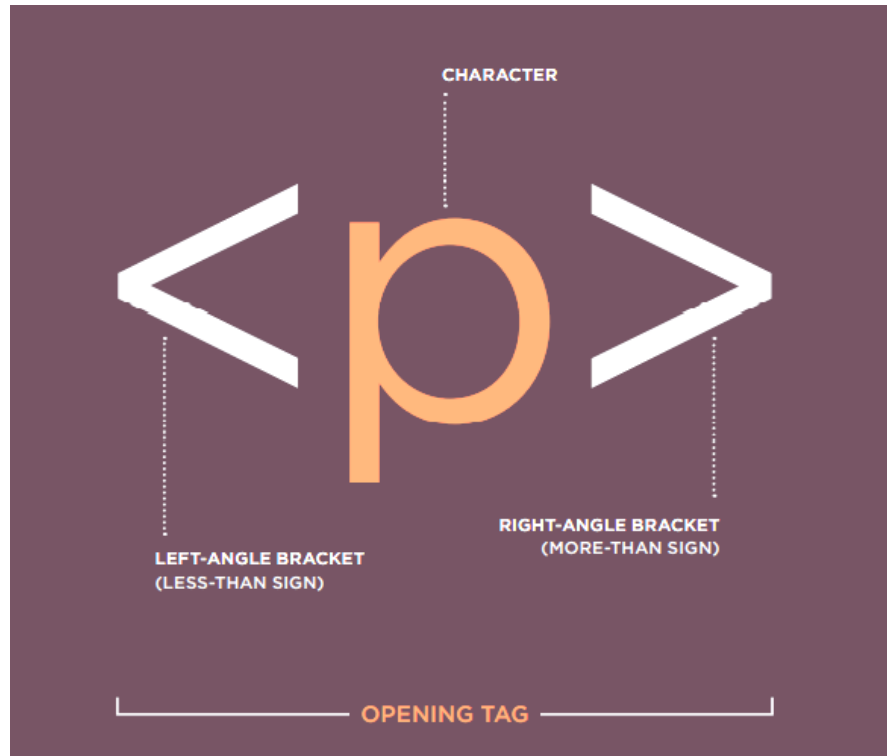
- What do you see in a newspaper? Stories
- Each Story has
 - Headline
 - Sub Headings
 - Paragraphs
 - Images
 - Quotes

The structure is very similar when a news story is viewed online (although it may also feature audio or video)

HTML Terminology

- HTML elements
 - Made up of characters that live inside angled brackets. Eg:<head>
 - Elements are usually made up of two **tags**: an opening tag and a closing tag. Eg:<head> </head>
 - Each HTML element tells the browser something about the information that sits between its opening and closing tags.

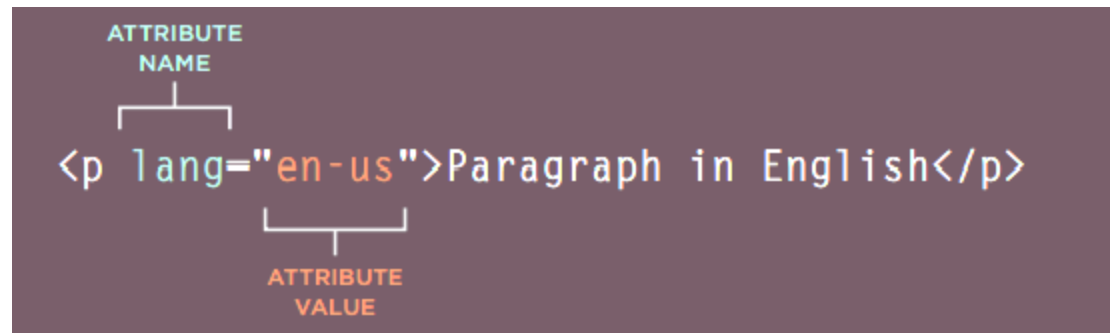
HTML Terminology



HTML Terminology

• Attributes

- Attributes provide additional information about the contents of an element.
- They appear on the opening tag of the element and are made up of two parts: a **name** and a **value**, separated by an equals sign.
- The attribute name indicates what kind of extra information you are supplying about the element's content. It should be written in lowercase.
- The value is the information or setting for the attribute. It should be placed in double quotes.



HTML Basic Page Structure

```
<!DOCTYPE html>
  <html>
    <head>
      <title>Page Title</title>
    </head>
    <body>
      <h1>My First Heading</h1>
      <p>My first paragraph.</p>
    </body>
  </html>
```

HTML Basic Page Structure

The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document

The `<html>` element is the root element of an HTML page

The `<head>` element contains meta information about the HTML page

The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)

The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

The `<h1>` element defines a large heading

The `<p>` element defines a paragraph

HTML Page

HTML Markup	Formatted Text
<pre> <!DOCTYPE html> <html> <head> <title>Page Title</title> </head> <body> <h1>My First Heading</h1> <p>My first paragraph.</p> </body> </html> </pre>	My First Heading My first paragraph.

HTML Meta data

- Metadata is data (information) about data.
- <meta> tags always go inside the <head> element, and are typically used to specify character set, page description, keywords, author of the document, and viewport settings.
- Metadata will not be displayed on the page, but is machine parsable.
- Metadata is used by browsers (how to display content or reload page), search engines (keywords), and other web services.
- Eg:
 - **Define keywords :** <meta name="keywords" content="HTML, CSS, JavaScript">
 - **Define a description :** <meta name="description" content="Free Web tutorials for HTML and CSS">
 - **Define the author:** <meta name="author" content="John Doe">
 - **Refresh document every 30 seconds:** <meta http-equiv="refresh" content="30">

HTML Comments

- HTML comments are not displayed in the browser, but they can help document your HTML source code.
- Comments are great for debugging HTML.
- Comments can be added to HTML source code by using the following syntax:

```
<!-- Write your comments here -->
```

```
<!-- This is a comment -->
```

```
<p>This is a paragraph.</p>
```

```
<!-- Remember to add more information here -->
```

HTML Headings

- HTML has six "levels" of headings: <h1>.....<h6>
- <h1> is used for main headings <h2> is used for subheadings.
- Browsers display the contents of headings at different sizes.

HTML Markup	Formatted Text
<pre><!DOCTYPE html> <html> <body> <h1>Heading 1</h1> <h2>Heading 2</h2> <h3>Heading 3</h3> <h4>Heading 4</h4> <h5>Heading 5</h5> <h6>Heading 6</h6> </body> </html></pre>	Heading 1 Heading 2 Heading 3 Heading 4 Heading 5 Heading 6

HTML Text Formatting Tags

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text
- `<p>` - Paragraph

HTML Semantic Tags

- There are some text elements that are not intended to affect the structure of your web pages, but they do add extra information to the pages.
- `<em>text</em>`
- `<strong>text</strong>`
- `<var>name</var>`
- `<dfn>a definition</dfn>`
- `<cite>a citation.</cite>`
- `<address>Janani.h@iit.ac.lk</address>`
- `<code>` Represents computer code `</code>`
- `<blockquote>text</blockquote>`
- `<abbr>IIT</abbr>`

Other Useful HTML tags

- Rulers
 - `<hr>`
- Paragraphs & Line Breaks
 - `<p>This is a parahraph</p>`
 - `<br>` Line Breaks

HTML List Tags

- HTML provides us with three different types:
 - Ordered lists are lists where each item in the list is numbered.
 - Eg: The list might be a set of steps for a recipe that must be performed in order, or a legal contract where each point needs to be identified by a section number.
 - Unordered lists are lists that begin with a bullet point (rather than characters that indicate order).
 - Definition lists are made up of a set of terms along with the definitions for each of those terms.

Ordered Lists

HTML Markup	Formatted Text
<pre><!DOCTYPE html> <html> <body> <h2>An ordered HTML list</h2> Coffee Tea Milk </body> </html></pre>	An ordered HTML list 1. Coffee 2. Tea 3. Milk

Ordered HTML List - The Type Attribute

The **type** attribute of the `<ol>` tag, defines the type of the list item marker:

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

Unordered Lists

HTML Markup	Formatted Text
<pre><!DOCTYPE html> <html> <body> <h2>An unordered HTML list</h2> Coffee Tea Milk </body> </html></pre>	An unordered HTML list • Coffee • Tea • Milk

Definition Lists

HTML Markup	Formatted Text
<pre><!DOCTYPE html> <html> <body> <h2>A Description List</h2> <dl> <dt>Coffee</dt> <dd>- black hot drink</dd> <dt>Milk</dt> <dd>- white cold drink</dd> </dl> </body> </html></pre>	A Description List Coffee - black hot drink Milk - white cold drink

Links

- Links are the defining feature of the web because they allow you to move from one web page to another.
- You will commonly come across the following types of links:
 - Links from one website to another
 - Links from one page to another on the same website
 - Links from one part of a web page to another part of the same page
 - Links that open in a new browser window
 - Links that start up your email program and address a new email to someone

HTML Links

- Links are created using the `<a>` element. Users can click on anything between the opening tag `<a>` and the closing `</a>` tag.
- You specify which page you want to link to using the href attribute.



HTML Link Types

- **Absolute links** - Takes the user to a specific page on the Web outside of the current website which contains the complete web URL.
 - E.g: “http://www.iit.ac.lk”
- **Relative links** - Refer to pages on the same website. Do not need to specify the domain name in the URL.
 - /courses
 - /graphics/image.png
- **Internal Links** - Accessing to different sections of the same page
- In addition, links can be created to download external files or Audio and video files.

HTML Links - The target Attribute

- By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.
- The target attribute specifies where to open the linked document and will have below values:
 - `_self` - Default. Opens the document in the same window/tab as it was clicked
 - `_blank` - Opens the document in a new window or tab
 - `_parent` - Opens the document in the parent frame
 - `_top` - Opens the document in the full body of the window

```
<a href="https://www.w3schools.com/" target="_blank">Visit W3Schools!</a>
```

HTML email Link

- Element <a> is used to create a link that starts up the user's email program and addresses an email to a specified email address,
- The value of the href attribute starts with mailto: and is followed by the email address you want the email to be sent to.

```
<a href="mailto:someone@example.com?Subject=Hello%20again" target="_top">Send Mail</a>
```

Review Questions

- Question 1 - Which HTML tags are introduced with HTML 5?
- Question 2 - Which HTML tags are used to create an ordered list?
- Question 3 - Write the HTML tags to display the following nested list.

Lists can be nested (list inside list):

1. Coffee
2. Tea
 1. Black tea
 2. Green tea
3. Milk

READING

- Duckett, J. (2011), *HTML & CSS, Design and Build Websites*, Wiley.
- Castro & Hyslop, B.(2011), *HTML5&CSS: Visual QuickStart Guide*, Peachpit Press.
- Robinson, J . (2018), *Learning Web Design, A Beginner's Guide to HTML,CSS, JavaScript, And Web Graphics*, O'Reilly.
- Online :
 - <http://www.w3schools.com/html/>
 - <https://www.udemy.com/learn-html5-programming-from-scratch/>